Foundations of Computer Programming: Naming conventions, Variables and Data Types

Learning objectives:

- Understanding naming conventions
- Understand the basic data types
- Understand the use of variables
- Understand the difference between mutable and immutable
- Practicing using the naming conventions, variables and data Types

Lesson exercise: Week 6

1. Group presentation: Naming conventions and PEP 8

Individual tasks:

2. The following a list of integers: numbers = [3, 7, 2, 9, 4, 1, 8]
   I. Append the number 5 to the list.
   II. Remove the number 9 from the list.
   III. Sort the list in ascending order.
   IV. Reverse the list after sorting it.
   V. Insert the number 6 at the index 2.
   VI. Print the index of the number 4 in the list.

3. The following tuple of mixed data types: my_tuple = (15, "Python", 3.14, True, 42, False)
   I. Print the index of the string "Python".
   II. Count how many times the value True appears in the tuple.
   III. Prints the elements from index 1 to index 4.

4. Create a python dictionary representing the marks of students (at least 15 students) in a test.
   I. Add a new student "Frank" with a mark of 60.
   II. Print all student names and their corresponding marks.

5. Explain mutable and immutable with examples.
6. Explain local and global variables in python with examples.